

Deploy digital twin with machine learning forecasts to optimize thermal energy across manufacturing facilities.

Manufacturing × Energy optimisation × Digital twin · OS093 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
46 is the world moving	50 can we play, can we win	MEDIUM 0 named in the evidence	€969m partial evidence · per year	LATER research and standards activi...	L1 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is to deploy a digital twin that fuses physics-based thermal models with machine-learning forecasts and multi-objective optimization to coordinate process heat, waste-heat recovery, thermal storage, and on-site renewables in real-time across manufacturing facilities. It targets manufacturers with significant thermal energy consumption and siloed energy management, and it exists now because recent research demonstrates the technical feasibility and benefits of such integrated systems.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€1.4bn €171m – €8.8bn	€969m €123m – €6.3bn	€11.6m €1.5m – €76.0m	partial
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€20.9m €20.9m – €56.6m	€251k €251k – €679k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Digital twin	10.2 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOTDPP	2021
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E_IOTDPP is used as a proxy for Digital twin: IoT in production as the observable precursor. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 2 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

Observed: tendered contract value, annualised

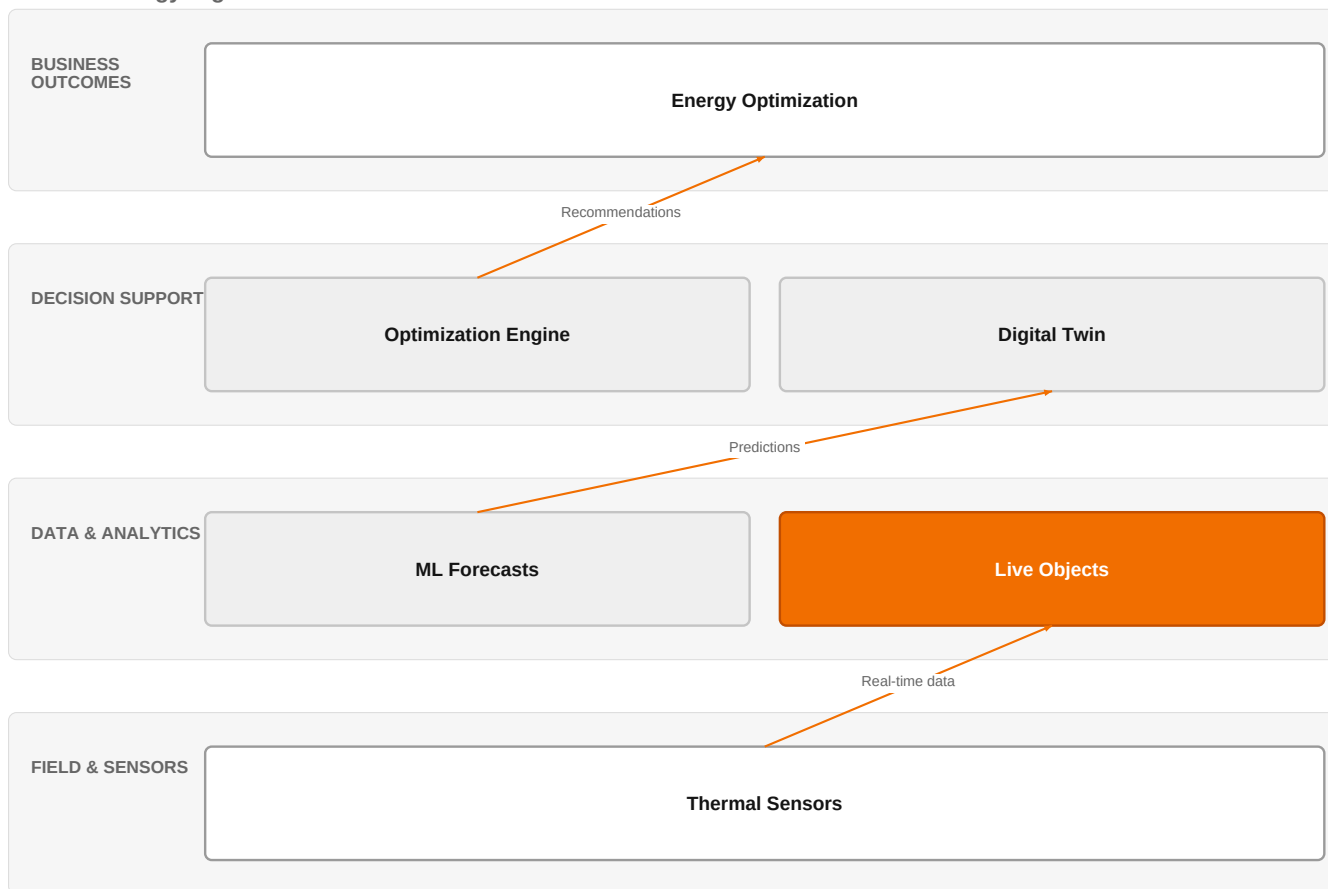
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a

market size. 1 of 2 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

The solution, in one picture

Thermal Energy Digital Twin Architecture



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how real-time thermal data flows through Orange's IoT platform into a digital twin that optimizes energy use across the facility.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Manufacturing processes consume substantial thermal energy, yet siloed management approaches cannot exploit facility-wide synergies. Recent research validates an integrated digital twin that fuses physics-based thermal models with machine-learning forecasts and multi-objective optimization to coordinate process heat, waste-heat recovery, thermal storage, and on-site renewables in real-time. Additionally, AI-driven digital twins are emerging as transformative tools in smart manufacturing, enabling real-time monitoring, predictive analytics, and autonomous decision-making. However, the high cost of data acquisition in industrial environments and the challenge of model degradation due to concept drift remain barriers to industrial uptake.

Sources: SIG-216688A4B953 [openalex.org 2026-03-01](https://openalex.org/2026-03-01); SIG-99692E33DB88 [openalex.org 2025-12-24](https://openalex.org/2025-12-24); SIG-AE9A0D62F13D [openalex.org 2026-04-01](https://openalex.org/2026-04-01); SIG-7AFE53FCEBE7 [arxiv.org 2026-07-20](https://arxiv.org/2026-07-20)

Who buys, and why now

The buyer is typically a plant operations director or energy manager at a manufacturing company with multiple facilities and significant thermal energy use. The pain is operational: energy costs are high, waste heat is not recovered, and thermal storage and renewables are not coordinated, leading to inefficiencies. The trigger is often a new sustainability target, rising energy prices, or a mandate to reduce carbon emissions, prompting a search for a facility-wide optimization solution.

Why it is hot now — every claim bound to a dated source

- Digital twin fuses physics-based thermal models with machine-learning forecasts and multi-objective optimization to coordinate process heat, waste-heat recovery, and thermal storage in real-time. [SIG-216688A4B953]

- Manufacturing processes consume substantial thermal energy, and siloed management approaches cannot exploit facility-wide synergies. [SIG-216688A4B953]
- An integrated digital twin fuses physics-based thermal models with machine-learning forecasts and multi-objective optimization to coordinate process heat, waste-heat recovery, thermal storage, and on-site renewables in real-time. [SIG-216688A4B953]
- A digital twin fusing physics-based thermal models with machine-learning forecasts coordinates process heat, waste-heat recovery, thermal storage, and on-site renewables in real-time. [SIG-216688A4B953]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble an engagement leveraging its IoT experts and Orange Group researchers to deploy a digital twin solution. The solution would use Live Objects as the IoT platform to collect real-time data from sensors across the facility. Orange would integrate physics-based thermal models with machine-learning forecasts and multi-objective optimization algorithms, drawing on its research capabilities. The deployment would follow a phased approach: first, assess current thermal energy flows; second, deploy sensors and connect to Live Objects; third, build and calibrate the digital twin; fourth, implement optimization recommendations; and finally, provide ongoing support and refinement.

Why Orange can win

Orange can win by combining its IoT expertise (Live Objects) with its research capabilities (Orange Group researchers) to deliver a sovereign, secure solution (ISO 27001, TISAX). Unlike pure software vendors, Orange can offer end-to-end integration, from connectivity to analytics, and can leverage its experience with reference customers in manufacturing. However, the assets are relatively thin for this specific use case; Orange lacks a dedicated digital twin platform or deep domain expertise in thermal engineering, so the win would depend on strong partnerships and the ability to integrate best-of-breed components.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer addresses the use case but not this technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 6.4; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
Accenture	Global systems integrator	sells Digital twin; active in Manufacturing; competes in OX: Smart Industries	structural	—
Capgemini	Global systems integrator	sells Digital twin; active in Manufacturing; competes in OX: Smart Industries	structural	—

Dassault Systemes / 3DS Outscale	Sovereign cloud provider	sells Digital twin; active in Manufacturing; competes in OX: Smart Industries	structural	—
PTC (ThingWorx)	Industrial / OT specialist	sells Digital twin; competes in OX: Smart Industries	structural	—
Schneider Electric	Industrial / OT specialist	sells Digital twin; active in Manufacturing; competes in OX: Smart Industries	structural	—
Siemens	Industrial / OT specialist	sells Digital twin; active in Manufacturing; competes in OX: Smart Industries	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	active in Manufacturing; competes in OX: Smart Industries	structural	—
Vodafone Business	Telecom operator peer	active in Manufacturing; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How do you currently manage thermal energy across your facilities, and is it siloed by process or centralized?
- 2 What is your biggest pain point: energy cost, carbon reduction targets, or operational inefficiency?
- 3 Do you have real-time sensor data available for process heat, waste-heat recovery, thermal storage, and on-site renewables?
- 4 Have you attempted any digital twin or energy optimization projects before? What were the outcomes?
- 5 Who would be the primary stakeholder for this initiative: plant operations, energy management, or sustainability?
- 6 What is your timeline for achieving energy optimization goals, and what budget is allocated?

Objections you will meet

They will say	You can answer
We already have a digital twin from Siemens or another vendor.	That's true, and they have strong platforms. However, our approach focuses on integrating physics-based thermal models with machine-learning forecasts and multi-objective optimization, which may not be fully covered by existing solutions. We can complement your current investments.
The cost of data acquisition is too high.	We understand that data acquisition can be expensive. Our approach leverages existing sensors where possible and uses advanced modeling to reduce the need for extensive instrumentation. We can start with a pilot to demonstrate value.
We are concerned about data security and sovereignty.	Security is a priority for us. Our solution is built on ISO 27001 certified infrastructure and can be deployed on sovereign cloud options, ensuring your data remains protected and compliant.

Risks and unknowns

The main risk is that the digital twin's machine learning models may degrade over time due to concept drift, requiring continual validation and updating. Additionally, the high cost of data acquisition in industrial environments could limit the accuracy of the twin. It is not yet known whether Orange can assemble the necessary domain expertise in thermal engineering or whether customers will accept a solution that relies on third-party components. The business case for energy savings may also be uncertain without a clear baseline.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a manufacturing prospect, leveraging Live Objects and ISO 27001 certification, and referencing Colgate-Palmolive and Saint-Gobain Glass as proof points for digital twin thermal energy optimisation.
Sales	In a conversation with a manufacturing operations director, say: 'We've helped manufacturers like Colgate-Palmolive and Saint-Gobain Glass use IoT and data analytics to improve operations, and we're exploring how digital twins can optimise thermal energy across facilities. Could we discuss your current energy management challenges?'
Strategist / Innovator	Commission a deep-dive brief on digital twin thermal energy optimisation for manufacturing, using Orange Group researchers and IoT experts to validate the technology stack and identify a pilot facility among existing manufacturing references.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-AE9A0D62F13D	2026-04-01	openalex.org	1	Quasi-Static Hypergraph Neural Networks: A High-Performance Approach for Digital Twin Modeling ...
SIG-216688A4B953	2026-03-01	openalex.org	1	Digital Twin-Enabled Thermal Energy Management System for Sustainable Manufacturing Process Opt...
SIG-505D0AD4A3FA	2026-02-13	openalex.org	1	Digital twin model for performance analysis and manufacturing process optimization
SIG-A067578BB3A4	2026-02-06	openalex.org	1	Optimization of the Design and Manufacturing Processes for Metal Additive Manufacturing Through...
SIG-99692E33DB88	2025-12-24	openalex.org	1	AI-Driven Digital Twins for Manufacturing: A Review Across Hierarchical Manufacturing System Le...
SIG-0964410DC575	2025-12-17	openalex.org	1	Generative and Predictive AI for digital twin systems in manufacturing
SIG-6C22EFB6A3A7	2025-12-17	openalex.org	1	On the reusability of machine learning-based process monitoring systems for manufacturing digit...
SIG-1CF55C438096	2025-10-15	openalex.org	1	Digital Twin for Automated Post-processing Chain in Additive Manufacturing
SIG-E4A301E44BE7	2025-08-26	openalex.org	1	Digital twin-driven decision support approach of production process for smart manufacturing
SIG-284B6A4B4E17	2025-08-19	openalex.org	1	Design and modeling of process manufacturing-oriented digital twin system
SIG-7785660FD9FA	2025-08-19	openalex.org	1	Application of Multimodel Concept and Virtual Assistants to Build a Digital Twin for Supporting...
SIG-288B4B508030	2026-08-10	arxiv.org	3	Automated generation of experimentally validated digital twins for desiccant-based low-dew-poin...
SIG-7AFE53FCEBE7	2026-07-20	arxiv.org	3	A Continual Validation, Updating, and Decision-Making Framework for Self-Adaptive Digital Twins...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:55+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.